

Development and exploratory testing of an analysis pipeline for ADC-derived radiomics biomarkers in brain tumor patients

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Abstract

Background and Motivation: This project aimed to develop a pipeline for extraction and subsequent analysis of radiomics features derived from apparent diffusion coefficient (ADC) maps. Exploratory testing of the pipeline was performed on a cohort of brain tumor patients, investigating the correlation of local treatment failure (LF) with radiomics features derived from pre-treatment ADC distributions within the gross tumor volume (GTV). *Materials and Methods:* A framework for rigid and affine multimodal image registration of ADC maps to computed tomography (CT) images was implemented alongside methods for harmonization of ADC maps. The patient cohort comprised 11 chordoma and chondrosarcoma patients treated with intensity-modulated proton therapy, with 3 patients exhibiting LF. For each patient, 53 radiomics features were extracted from the ADC map of the GTV using pre-implemented methods. Spearman rank correlation coefficients were computed to assess the association between each feature and LF, with statistical significance evaluated via permutation testing. *Results:* Exploratory analysis showed that overall raw ADC values were significantly elevated within the tumor volume. A broader trend of increased heterogeneity in the tumor was observed to correlate positively with LF; however, no individual radiomics feature reached statistical significance. *Conclusion:* The developed pipeline offers a reproducible method for extracting ADC-derived radiomics features. Although no statistically significant associations with LF were found in this exploratory cohort, the observed trends warrant further investigation of the prognostic potential of ADC-based radiomics in brain tumor outcomes.

1 Introduction

1.1 Background and Motivation

According to the World Health Organization, cancer accounted for 1 in 6 deaths globally as of 2018, with numbers projected to further increase over the coming decades [1]. Therefore, research and innovation in cancer prevention, diagnosis and treatment is a major societal priority. Radiotherapy (RT) is an important treatment modality, as approximately one in two patients receive a form of RT during their treatment [2, 3]. In external beam RT, localized high doses of radiation are deposited on the tumor to induce irreversible damage to cancer cells, inhibiting their spread. While conventional RT delivers dose through x-ray photons, an increasing number of patients are being treated with charged particles therapy, using protons or heavier ions like helium (^4He) or carbon (^{12}C). Since charged particles deposit their energy more selectively than photons, particle therapy has the potential to reduce radiation dose to healthy tissues, which may lower the risk of radiation-induced adverse side effects [4, 5, 6]. The

higher precision of particle therapy enables targeted dose escalation, or dose painting, based on individual patient response. However, despite the advantages offered by PT, some patients still experience relapses after termination of their treatment. Prediction of treatment outcome is therefore of great importance and predictive models may be aided by the incorporation of a plethora of information obtained from medical images taken before and during the treatment process.

Radiotherapy is inherently an image-guided intervention, typically planned on Computed Tomography (CT) images and often complemented by functional and anatomical information from Magnetic Resonance (MR) imaging. While CT provides the tissue electron density required for dose calculation and treatment planning, MR brings high soft-tissue contrast for precise tumor localization and functional insights into the tissue microenvironment. Among others, Diffusion-Weighted Imaging (DWI) provides such functional information with a wide range of applications in RT, from tumor delineation and cancer staging, to early response monitoring and post-treatment response evaluation [7, 8, 9]. Most importantly in outcome modeling, quantitative image biomarkers (QIBs) can be extracted from baseline DW images.

However, the integration of DWI into clinical data analysis requires dedicated image processing to correct for geometric distortions and to accurately map DWI-derived QIBs onto the treatment planning geometry and patient anatomy.

1.2 ADC and Radiomics in Outcome Analysis

DWI is an MR-based imaging technique, which uses two pulsed field gradients to sensitize the signal to water diffusion. The first gradient pulse dephases proton spins, and the second inverse pulse attempts to rephase them. If no diffusion occurs between the two pulses, rephasing is exact and no signal loss is observed. However, in regions with high diffusion imperfect rephasing leads to signal attenuation. The ratio between the signal $S(TE)$ with diffusion-sensitizing gradients and the signal S_0 without can be calculated as

$$\frac{S(TE)}{S_0} = \exp \left[-\gamma^2 G^2 \delta^2 \left(\Delta - \frac{\delta}{3} \right) D \right] \quad (1)$$

where γ [$\text{s}^{-1}\text{T}^{-1}$] is the gyromagnetic ratio, G [T/m] is the diffusion sensitizing pulse's strength and δ [s] is its duration, Δ [s] is the time between the dephasing and rephasing pulse and D [m^2/s] is the diffusion coefficient [10]. The measurement of signal loss is however complicated by the presence of the gradient fields used for signal localization in MRI, leading to deviations from the behavior predicted by Equation 1. Hence a simplified relation for $S(TE)/S_0$ has been proposed [11]:

$$\frac{S(TE)}{S_0} = \exp(-b \cdot \text{ADC}) \quad (2)$$

The b -value [s/mm^2] summarizes the effect of all gradient fields, controls the strength of diffusion weighting and is dependent on user-controlled acquisition parameters. By acquiring diffusion-weighted images at multiple b -values, the apparent diffusion coefficient (ADC) [mm^2/s] value can be estimated at the voxel level, summarizing the various diffusion contributions within the underlying tissue in a single value. ADC maps reflect tissue characteristics related to microstructural cellular organization and density. Specific aspect of this information

on tissue characteristics may be investigated further and condensed by extracting so-called radiomics features from ADC maps. Features may range from standard statistical measures such as skewness and kurtosis to measures of intensity level heterogeneity and tissue shape. Whether used as raw values or processed by extracting radiomics features, such DWI-based information has shown promise in predicting tumor histologic grade and treatment outcomes across various tumor types and therapeutic approaches [12, 13, 14, 15, 16].

In practice, the extraction of DWI-based radiomics features is frequently limited to a pre-defined region of interest (ROI), such as the tumor volume. Since ROIs are generally delineated on CT images, an alignment process of the ADC map and CT image, referred to as image registration, is necessary. Registration is complicated by the fact that CT images hold anatomical information, whereas ADC maps are functional images with no direct anatomical information. To bridge this gap, an intermediate multi-modal registration with a T1- or T2-weighted MR image is usually performed. This facilitates the process, since ADC, T1 and T2 images are MR-based and may be acquired during the same imaging session, while T1 and T2 images also hold anatomical information, making them suitable for registration with CT images.

1.3 Project Objectives

This study aims to develop and evaluate a processing pipeline for analyzing radiomics features derived from apparent diffusion coefficient (ADC) maps in skull base tumor patients treated with proton therapy. The pipeline integrates multi-modal image registration, tumor volume reconstruction, and ADC normalization to enable consistent extraction of radiomics features. A key focus is on addressing the challenge of mapping ADC values onto planning CT images in the absence of corresponding T2-weighted MRI or low b -value DWI, which are commonly used intermediates for registration, but often lacking in retrospective studies. To demonstrate the pipeline’s potential, it is applied to a small dataset, with exploratory analyses of ADC distributions across tumor regions and correlations between radiomics features and local treatment failure. The overarching goal is to establish a reproducible framework to facilitate investigations into ADC-based predictors of radiotherapy outcomes.

2 Image Registration and Processing Pipeline

2.1 Patient Dataset

This retrospective study was conducted on a cohort of 11 chordoma and chondrosarcoma cancer patients treated with proton therapy between 2015 and 2021 at the Paul Scherrer Institute. Among these, 3 experienced local (in-field) treatment failure. The analysis was based on pre-treatment baseline ADC maps, T1-weighted MR images, planning CT scans and associated treatment volumes and regions of interest (ROI), including the gross tumor volume (GTV).

Diffusion-weighted images were acquired using a sequence based on the RESOLVE technique (Readout Segmentation Of Long Variable Echo-trains), which reduces susceptibility artifacts and improves spatial resolution compared to conventional single-shot echo-planar imaging. Imaging was performed on a 1.5 T MAGNETOM Aera MRI system (Siemens Healthineers, Erlangen, Germany). ADC maps were calculated using a mono-exponential model from two

b -values (0 and 1000 s/mm²), according to Equation 2.

T1-weighted images were acquired using a variety of clinical protocols across subjects. The majority of subjects underwent a 3D MPRAGE sequence (tf3d1_16), either in native form (e.g., t1_mprage_tra_nativ) or with post-contrast fusion (e.g., t1_mprage_tra_fusion, T1 mprage tra fusion nach KM). One subject was scanned using a 3D VIBE Dixon sequence (f3d2) and three subjects with a 2D TSE sequence (tse2d1rr4, t1_tse_tra_3mm). These variations reflect standard clinical imaging practices.

Planning CT images were uniformly acquired on a SOMATOM scanner (Siemens Healthineers, Erlangen, Germany) operating at 120 kVp in helical mode.

Data were accessed in anonymized form, and their use was approved by the ethics committee Ethikkommission Nordwest- und Zentralschweiz under the agreement EKNZ-2023-01826.

2.2 Tumor Volume Reconstruction

Due to multiple manipulations of the planning images for use in a historical treatment planning system (PSIPlan) applied to this cohort, the clinical treatment volumes were available only at coarse resolution. Therefore, mesh interpolation has been applied to restore the GTV contours at original planning CT slice thickness. This involved first connecting contour points on each axial slice, and their resampling to a fixed number of points. Contour slices at finer resolution were then obtained by slicing the 3D mesh of the GTV obtained by Delaunay triangulation of the resampled contour points. After having populated all slices of the original planning CT data and smoothed out nonnatural points (due to triangulation artifacts), masks could be created on each axial slice by generating polygons from the newly obtained contour points.

2.3 Image Registration Framework

The fusion of functional information derived from DWI with anatomical data from the planning CT and associated treatment volumes requires multi-modal image registration. However, direct registration is not feasible due to the differing image contrasts and information content of the two modalities, which hinder reliable image-based alignment. Instead, the registration process was split into multiple stages, first registering ADC to T1 and subsequently registering T1 to CT. The spatial transforms calculated in the two stages could then be concatenated to obtain the mapping from ADC to CT. While the registration process could also have been inverted, in clinical practice planning CT images are at the center of informational networks. Segmentation and dose information may be either directly based on the CT or be moved over from anatomical MR images via MR to CT registration. The choice to register ADC onto CT was therefore a practical one, necessitating the least amount of information transfer.

As with any other optimization problem, successful registration is dependent on a good initial guess, especially in cases with large setup variability at the scanner, such as in head and neck cancer patients. To obtain initial transforms, brain masks were extracted from the ADC, T1 and CT images to estimate a rough alignment. For the CT and T1 images, brain masks were extracted using TotalSegmentator, an open-source, AI-powered medical image segmentation tool [17]. More challenging are the ADC maps, which do not contain anatomical information per se, but instead represent a quantitative functional MRI biomarker. In this case, ADC

distributions were first clipped by defining a lower and upper cutoff value, $(0.3 \text{ and } 1.5) \times 10^{-3} \text{mm}^2/\text{s}$, respectively, to exclude background noise and include the majority of brain tissue. The values within this range were then linearly scaled to the interval $[0, 1]$, yielding soft masks of ADC maps that were inverted by setting the voxels of intensity $i > 0$ to $1 - i$ and smoothed with a gaussian filter with $\sigma = 0.5$. This resulted in a synthetic brain image that primarily represents white and gray matter, with an appearance comparable to that of an anatomical T1-weighted sequence. Binary maps were extracted from inverted soft masks by thresholding at 0.4 [normalized a.u.]. From that, binary brain masks were extracted by identifying the largest connected region and performing a 2-iteration closing morphological filtering to close holes using 3D kernels (18-connected neighborhoods).

Initial guess for the T1-CT registration stage were obtained by registering the signed distance fields (SDF) of the corresponding binary brain masks. Instead, due to the limited anatomical fidelity of the ADC-derived images, initial guesses for the ADC-T1 registration stage were obtained by aligning the centers of mass of the ADC and T1 brain masks. These were then used as starting point for the T1-CT and ADC-T1 registration based on multi-resolution normalized mutual information. Both rigid and affine registration was used at the CT-T1 registration stage. Importantly the above-explained inverted ADC soft masks were used as surrogate for the ADC maps in the ADC-T1 registration due to their similarity with T1 images. The parameters used for the registration are summarized in Table 1. Finally, the optimized transforms between the three images were concatenated, and the ADC maps resampled onto the planning CT grid using trilinear interpolation. The registration pipeline was implemented using *SimpleITK* Python API [18] and is presented schematically in Figure 1.

Table 1: Parameters of the registration methods forming the stages of the ADC to CT registration pipeline.

Registration Stage	T1-CT brain mask SDFs	ADC-T1	T1-CT
Transformation Type	Euler3D		Euler3D/Affine
Initial Estimate	Moments-bases alignment of brain mask SDFs	Moments-based alignment of brain masks	Optimized alignment of brain mask SDFs
Similarity Metric	MeanSquares	MattesMutualInformation (numberOfHistogramBins = 32)	
Metric Fixed Mask	None	T1 brain mask	CT head mask
Optimizer	RegularStepGradientDescent (learningRate = 1.0, minStep = 1e-4, numberOfIterations = 200, gradientMagnitudeTolerance = 1e-6)		
Multiresolution	None	ShrinkFactors = (4,2,1), SmoothingSigmas = (2,1,0)	
Interpolator	linear		

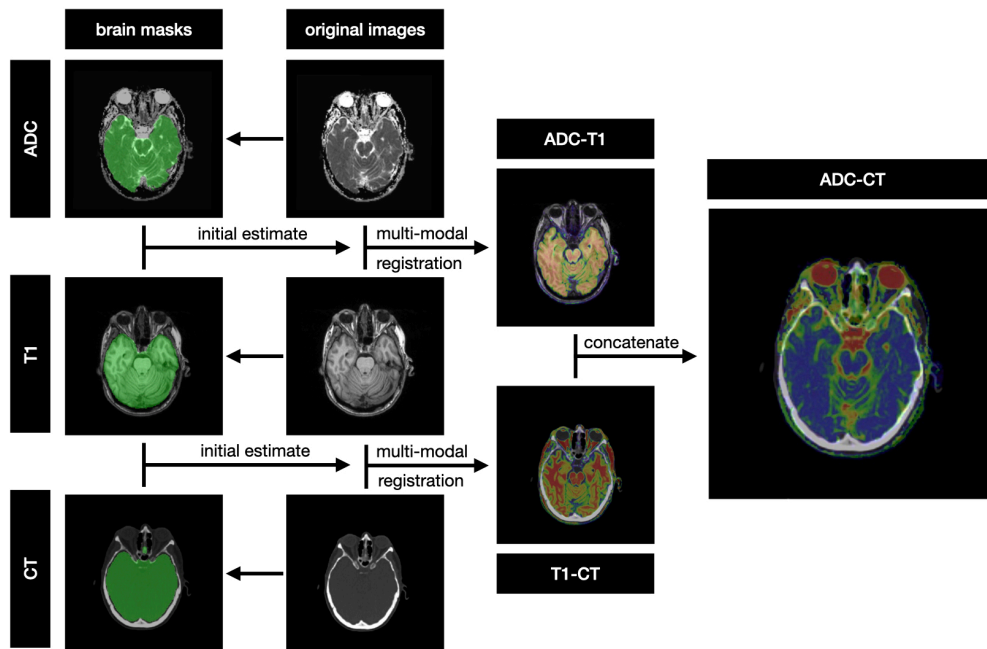


Figure 1: Stages of the ADC to CT registration pipeline. Importantly the represented ADC image is converted to an anatomical-alike surrogate before extraction of the binary brain mask and registration with T1.

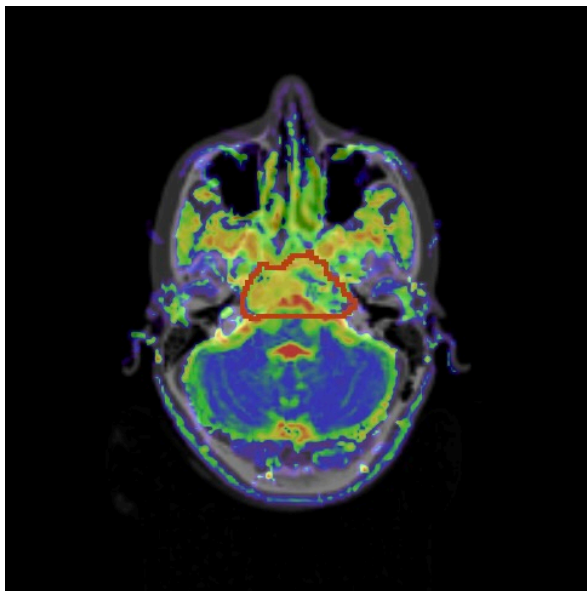


Figure 2: Figure showing registered ADC map of patient P0185 overlaid on planning CT image with GTV delineated in red.

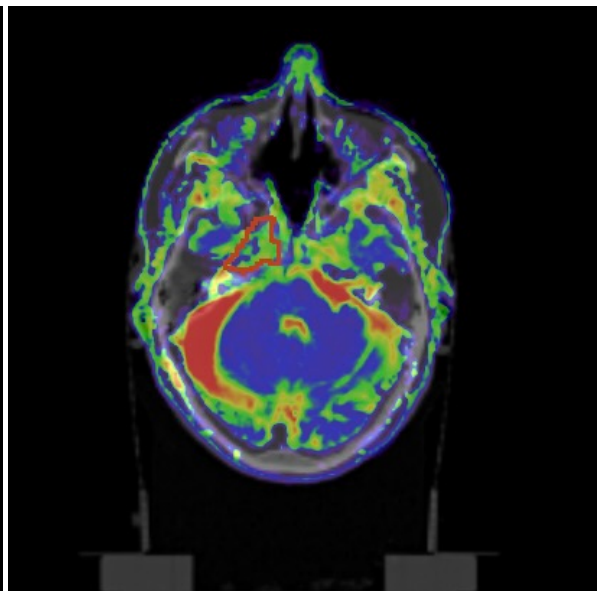


Figure 3: Figure showing registered ADC map of patient P0201 overlaid on planning CT image with GTV delineated in red.

3 Exploratory Analysis of Tumor ADC Distributions

In the following, ADC value distributions are compared within each patient and across the cohort, considering two volumes of interest: the GTV, as delineated by clinicians, and the brain outside the GTV segmented using TotalSegmentator, referred to as Brain in the following.

3.1 Impact of Registration Approach

The ADC distributions within GTV and Brain structures, obtained following rigid and affine registration for ADC-to-CT mapping, were compared across the patient cohort as shown in Figures 4 and 5 to assess the added value of incorporating scale and shear deformations. Qualitatively, the distributions appear very similar, though small shifts in the median, mean, IQR and extrema are noticeable in some patients. Two-sample Kolmogorov–Smirnov tests (KS-tests) were conducted to assess the significance of these differences.

They were found to be significant for Brain distributions in 5 patients ($p \leq 0.05$) and for GTV distributions in 3 patients. Based on this finding the following analysis was conducted for both registration methods, to assess how further processing steps would be affected by the choice of registration method.

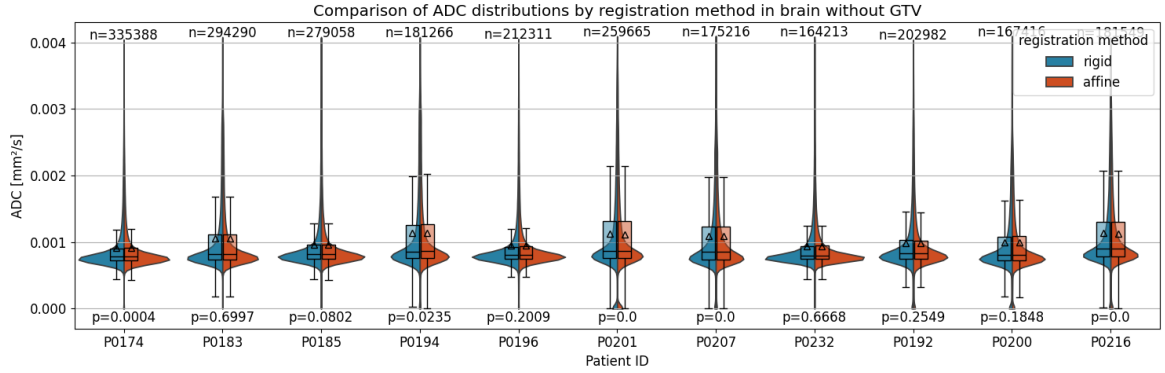


Figure 4: ADC distributions in the brain without GTV for all patients, obtained following rigid and affine registration. p -values correspond to paired KS-tests between distributions. n represents the number of data points in the distributions.

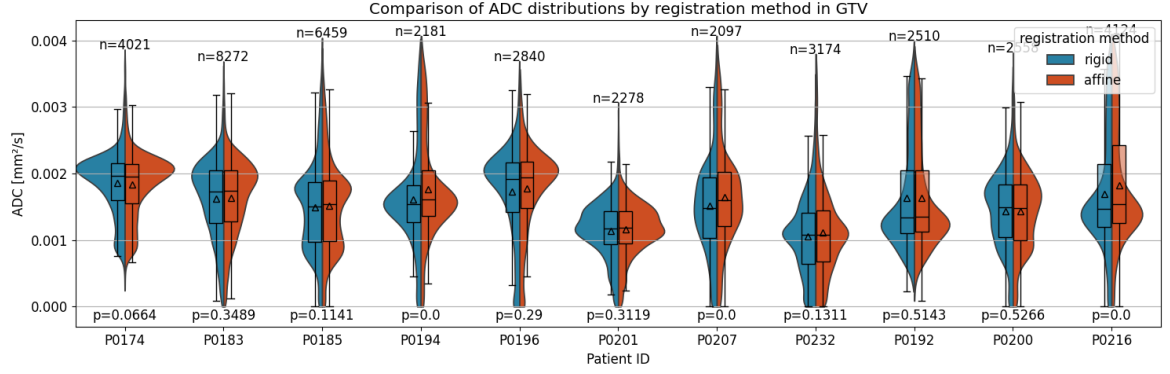


Figure 5: ADC distributions in the GTVs for all patients, obtained following rigid and affine registration. p -values correspond to paired KS-tests between distributions. n represents the number of data points in the distributions.

3.2 Regional ADC comparison

Side by side visual comparisons of the distributions of ADC values in both ROIs are presented for both registration methods in Figure 6 (rigid) and 7 (affine).

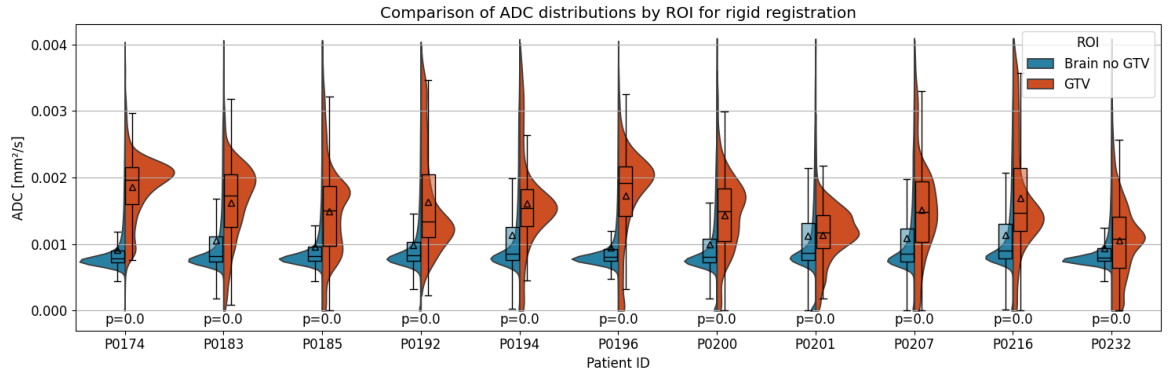


Figure 6: ADC distributions in the GTVs and brain without GTV for all patients, obtained following rigid registration. p -values correspond to paired KS-tests between distributions.

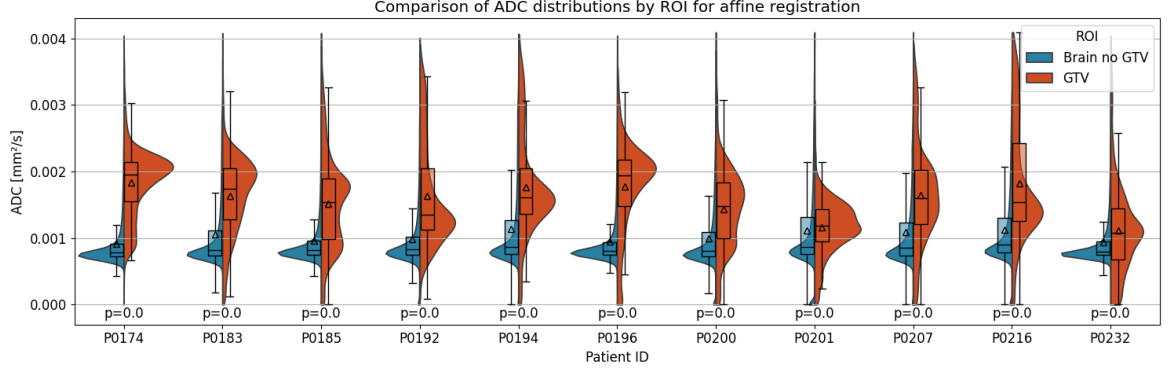


Figure 7: ADC distributions in the GTVs and brain without GTV for all patients, obtained following affine registration. p -values correspond to paired KS-tests between distributions.

The distributions in GTVs exhibit significant tails, and their flatness (kurtosis) is patient-specific. They reflect a general elevation of ADC in the GTV compared to Brain at the individual patient level, confirmed by statistically significant KS tests for difference in distributions. Descriptive statistics are presented in Table 2. Further analysis is provided in Section 3.3.

Table 2: Median and mean ADC values across the population for rigid and affine registrations in GTV and Brain.

Region	Statistic	Rigid ($\times 10^{-3}$ mm ² /s)	Affine ($\times 10^{-3}$ mm ² /s)
GTV	Median	1.5443	1.5775
	Mean	1.5495	1.5926
Brain	Median	0.8215	0.8218
	Mean	1.0192	1.0178

3.3 ADC Comparison by Treatment Outcome

To assess their prognostic relevance, ADC distributions in the GTV were compared across treatment outcome groups. Due to session specific bias [19], ADC maps and GTV structure masks were first resampled to a common voxel spacing of 1 mm $\times$ 1 mm $\times$ 2 mm using trilinear interpolation and ADC values were subsequently normalized to the individual patient Brain using robust z-scoring as follows:

$$\text{ADC}_{\text{normalized}} = 0.6745 \cdot \frac{\text{ADC} - \text{ADC}_{\text{median}}}{\text{MAD}} \quad (3)$$

where $\text{ADC}_{\text{median}}$ is the median ADC in the Brain and $\text{MAD} = \text{Median}(|\text{ADC} - \text{ADC}_{\text{median}}|)$ is the median absolute deviation of the ADC in the Brain. ADC distributions within the GTV resulting from the resampling and normalization process are displayed in Figure 8, with p -values from two-sample KS tests comparing the distributions obtained from rigid and affine registration. Of the 11 patients, 9 show significant differences between distributions obtained from the two registration methods.

Characteristic values of normalized ADC distributions distinguished by treatment outcome are listed in Tables 3 and 4 for ADC maps obtained from rigid and affine registration respectively. For both registration methods distributions with local treatment failure (LF) exhibit a decreased median ADC, elevated skewness and lower kurtosis.

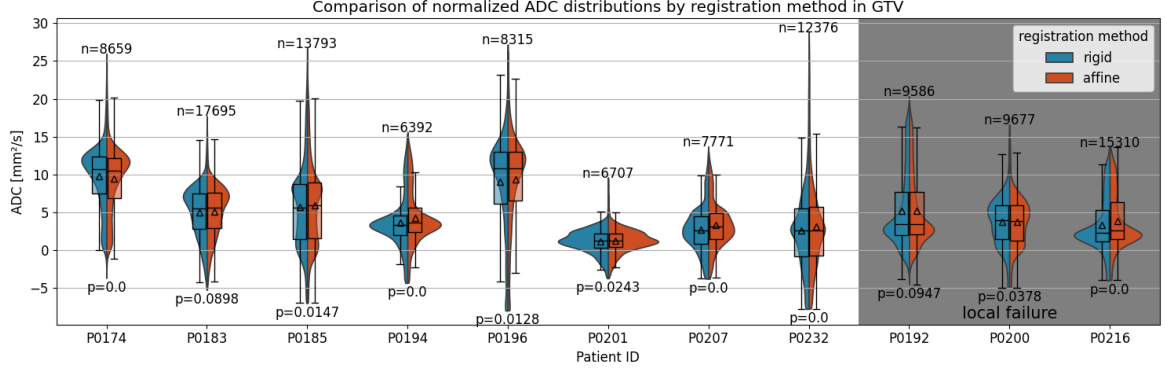


Figure 8: Normalized ADC distributions in the GTV after rigid and affine registration, resampling and robust z-score normalization. p -values correspond to paired KS-tests between distributions. n represents the number of data points in the distributions.

Table 3: Characteristic values of normalized ADC distributions in the GTV obtained from rigid registration, distinguished by treatment outcome.

	Median ADC		Skewness ADC		Kurtosis ADC	
	no LF	LF	no LF	LF	no LF	LF
Min	1.2566	2.3635	-0.9007	0.0119	-0.0612	0.0404
Max	10.8081	3.9965	0.6798	1.0268	1.1774	0.4401
Median	4.4393	3.4068	0.062	0.6518	0.5388	0.1311
IQR	4.2994	0.8165	0.8855	0.5074	0.3785	0.1999

Table 4: Characteristic values of normalized ADC distributions in the GTV obtained from affine registration, distinguished by treatment outcome.

	Median ADC		Skewness ADC		Kurtosis ADC	
	no LF	LF	no LF	LF	no LF	LF
Min	1.2777	2.6726	-0.865	0.1146	0.0311	-0.3086
Max	10.8587	3.8937	1.1343	1.0076	2.3653	0.4713
Median	4.6356	3.4969	0.1585	0.6239	0.4554	0.1172
IQR	4.0956	0.6105	0.8353	0.4465	0.7281	0.3899

4 Radiomics Feature Extraction and Outcome Correlation

The fusion of DWI-based maps with treatment planning information, including CT imaging and treatment contours, opens the possibility for radiomics studies on the ADC biomarker.

The following sections present an exploratory study aimed at correlating a set of standard radiomics features with local treatment failure.

4.1 Radiomics Feature Set and Extraction Method

Features were extracted from resampled, normalized ADC maps within GTVs using *Pyradiomics*, a widely used and validated radiomics package in oncologic imaging [20]. A fixed bin number approach with 128 bins was chosen for intensity discretization, as this approach was shown to yield better stability of radiomics features than a fixed bin size approach [21]. A total of 53 features were extracted for each patient, excluding directionally dependent features (features related to the Gray Level Co-occurrence Matrix and the Gray Level Run Length Matrix) as well as shape features, which describe the GTV geometry rather than the biomarker distribution.

4.2 Statistical Analysis and Correlation with Outcome

Each extracted radiomics feature was correlated with local failure using Spearman’s rank correlation coefficient (ρ), a non-parametric test. Due to the small size of the patient cohort included in this study, p -values associated with the correlation coefficients were calculated by permutation tests ($n = 10000$). The 95% confidence intervals for the correlation coefficients were extracted from distributions generated through bootstrapping, with $n = 10000$ samples each. Additionally, the values of radiomics features obtained from rigid and affine registration were compared by calculating two-way mixed single score intraclass correlation coefficients (ICC) for each feature using the *Pingouin* Python package [22].

ICC scores for comparison of feature values derived from rigid and affine registration are shown in Figure 9. Figures 10 and 11 show the results from the correlation analysis of radiomics features and local failure as derived from both registration modalities.

Several radiomics features exhibited correlations with local failure, although none reached statistical significance at the conventional $p < 0.05$ threshold in our cohort. Table 5 summarizes the most notable features and their correlation coefficients (ρ) and p -values under rigid and affine registration.

Table 5: Notable radiomics features correlated with local failure.

Feature	ρ_{rigid}	p_{rigid}	ρ_{affine}	p_{affine}
Firstorder_Kurtosis	−0.45	0.19	−0.39	0.28
GLSZM_GrayLevelNonUniformityNormalized	−0.39	0.28	−0.39	0.28
Firstorder_Uniformity	−0.32	0.31	−0.27	0.38
GLDM_SmallDependenceLowGrayLevelEmphasis	−0.30	0.34	−0.26	0.42
GLDM_GrayLevelVariance	0.52	0.14	0.45	0.18
Firstorder_Entropy	0.52	0.15	0.39	0.29
NGTDM_Contrast	0.45	0.20	0.39	0.28

Firstorder_Kurtosis and GLSZM_GrayLevelNonUniformityNormalized showed moderate inverse relationships, suggesting that tumors with more peaked intensity distributions and greater texture uniformity may be less prone to recurrence compared to more heterogeneous

distributions. Other features such as `Firstorder_Uniformity` and `GLDM_SmallDependenceLowGrayLevelEmphasis` followed similar trends, though with smaller effect sizes.

Conversely, several features demonstrated positive correlations with local failure, including `GLDM_GrayLevelVariance`, `Firstorder_Entropy`, and `NGTDM_Contrast`, suggesting that increased texture heterogeneity and disordered intensity patterns may be associated with a higher risk of local recurrence.

5 Discussion and Limitations

Comparability of ADC distributions obtained following rigid and affine registration varies across patients and analysis steps. Based on evaluations conducted shortly after the conclusion of this project it does however seem likely that the cases with poor comparability are a consequence of the choice of test used to assess the differences between distributions. This is supported by the good comparability of radiomics features between registration methods as well as the apparent visual similarity of distributions in all cases.

In our cohort of chordoma and chondrosarcoma cancer patients, GTV exhibited elevated ADC values compared to healthy brain tissue on baseline imaging. However, absolute comparison with values reported in the literature is challenging, as most prior studies stratify patients by tumor phenotype and pathological grade. Such distinction was not feasible in our cohort due to missing clinical data. Furthermore, ADC measurements are known to vary with acquisition parameters [23, 24], further complicating cross-study comparison. As a result, direct interpretation of absolute ADC distributions may be of limited use. A more meaningful approach involves normalization strategies, such as those employed in this study, or the analysis of ratios relative to reference tissue, typically healthy white matter in the context of brain tumors.

With due reservations, it can nonetheless be stated that our findings for mean and median ADC within tumor tissue are consistent with literature values of averages in low grade tumors. Hassannejad et al. reported an average ADC of $(1.5445 \pm 0.08131) \times 10^{-3} \text{mm}^2/\text{s}$ in Grade I adult brain tumors [25], which aligns with our observation. Conversely, our measured mean ADC values in the Brain were higher than those reported by Helenius et al., who found a mean ADC of $(0.89 \pm 0.04) \times 10^{-3} \text{mm}^2/\text{s}$. The elevated values observed in our study may be attributed to the inclusion of tumor adjacent tissue at the GTV boundary, potentially skewing the Brain assessment, as well as differences in statistical analysis.

Although not reaching statistical significance in our cohort, the radiomics analysis suggests a broader trend in which intra-tumoral heterogeneity, as quantified through ADC maps features, may carry prognostic value. However, this hypothesis requires validation in larger, independent datasets. A similar observation was reported by Jajodia et al. [16] for cervical cancer, although the specific features associated with treatment outcomes in that study differed from those found to be of interest in our analysis. Similarly, Bollen et al. found that `gldm_GrayLevelVariance` was significantly associated with locoregional control in oropharyngeal carcinoma [26]. As noted by Traverso et al. [27], features derived from GLSZM matrices often show limited reproducibility and conclusions should therefore be interpreted with caution.

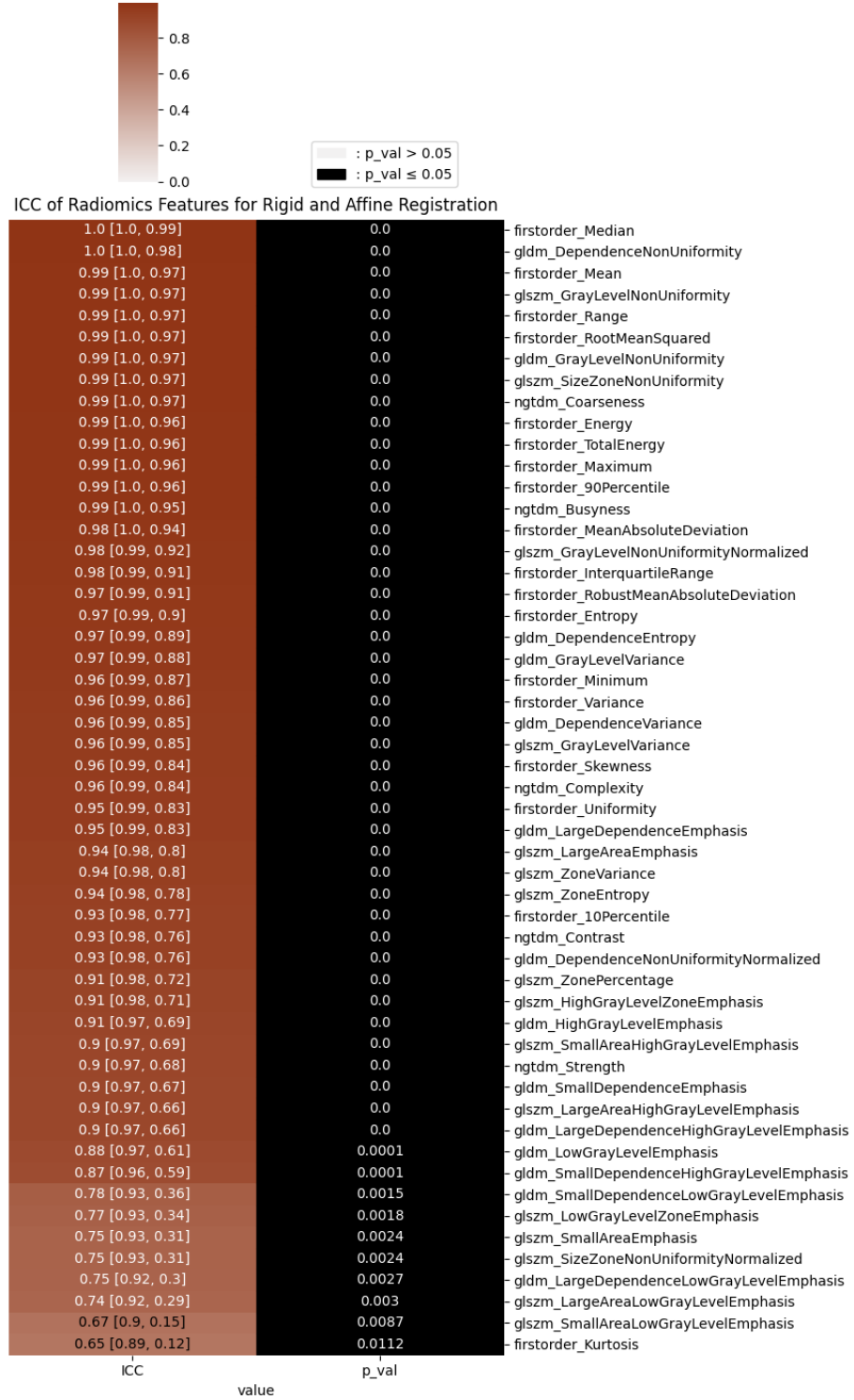


Figure 9: Intraclass correlation coefficients for all 53 extracted radiomics features, along with associated 95% confidence intervals and p-values.

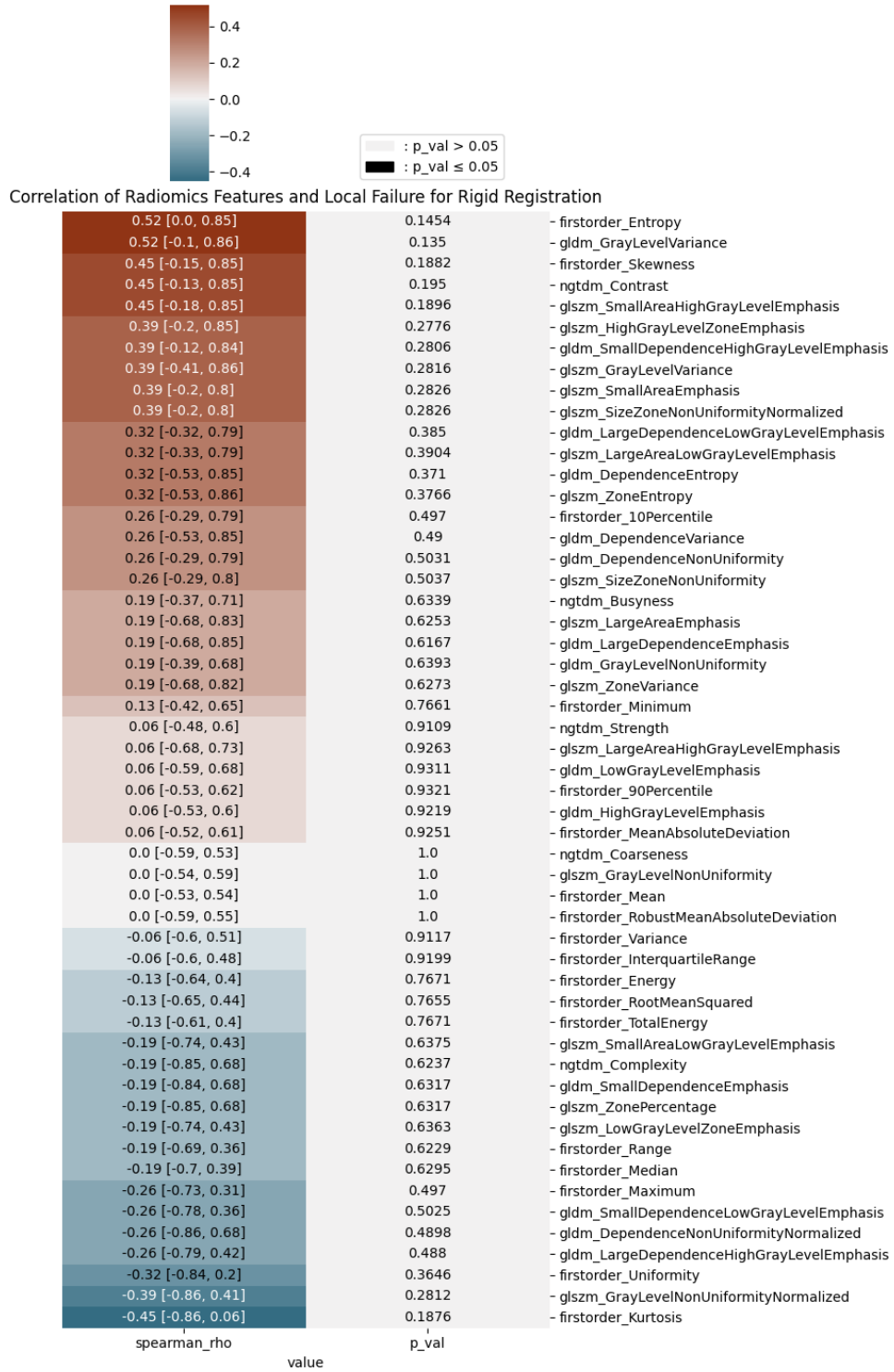


Figure 10: Spearman's rank correlation coefficient of 53 radiomics features with local treatment failure as derived from normalized ADC maps obtained by rigid registration. P-values associated with the correlation coefficients were calculated using permutation tests with $n = 10000$ permutations each, while 95% confidence intervals were extracted from 10000-sample distributions generated by bootstrapping.

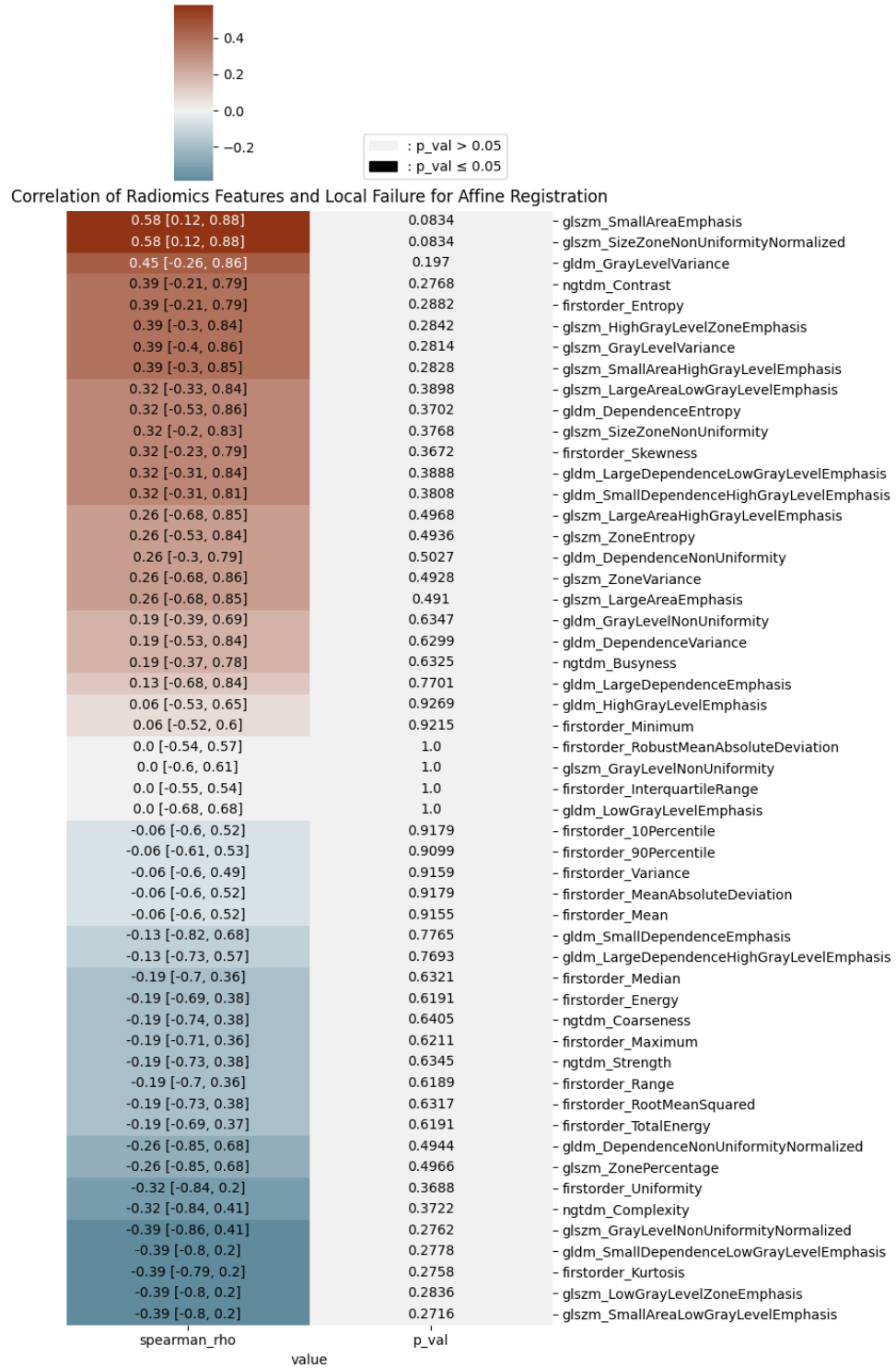


Figure 11: Spearman's rank correlation coefficient of 53 radiomics features with local treatment failure as derived from normalized ADC maps obtained by affine registration. P-values associated with the correlation coefficients were calculated using permutation tests with $n = 10000$ permutations each, while 95% confidence intervals were extracted from 10000-sample distributions generated by bootstrapping.

While the study is methodologically robust, several limitations remain, particularly concerning the exploratory analysis of clinical data. First and foremost is the small size of the patient cohort and the substantial imbalance in treatment outcomes, which limits the statistical power of the results. This is reflected in the large 95% confidence intervals and the overall lack of significant p-values in the radiomics correlations.

Secondly, concerns exist regarding the reproducibility of our radiomics findings. Features extracted from MRI-based images have been shown to be sensitive to various factors, including acquisition parameters, segmentation methods, normalization strategies, and intensity discretization [21, 27, 28, 29]. In the absence of a common standard for harmonizing MRI-based images, the specific pre-processing methods used in this project represents just one of several reasonable options.

Future studies on this topic would benefit from addressing these limitations by employing larger, more balanced patient cohorts, systematically evaluating alternative harmonization techniques, and incorporating more advanced multivariate regression schemes to account for potential confounders.

6 Conclusion

This study presents a reproducible image processing pipeline for mapping and analyzing radiomics features derived from diffusion-weighted MRI, specifically ADC maps, in brain tumor patients treated with proton therapy. By addressing challenges in multi-modal image registration and ADC normalization, the proposed framework enables integration of functional imaging biomarkers into the radiotherapy planning geometry. Application of the pipeline to a small retrospective cohort demonstrated overall elevated ADC values in tumor tissue and identified several radiomic features that showed moderate, though non-significant, correlations with treatment outcome. These preliminary findings support the potential of ADC-based radiomics in capturing intra-tumoral heterogeneity relevant to prognosis. However, limitations related to sample size, outcome imbalance, and feature reproducibility highlight the need for validation in larger, harmonized cohorts. The presented methodology lays the groundwork for future studies exploring the prognostic utility of ADC and advancing imaging-based personalization of proton therapy.

Acknowledgments

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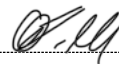
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